

ISLAY PORT ELLEN, United Kingdom

WMO: 031050

Lat: 55.683N

Lon: 6.250W

Elev: 17

StdP: 101.12

Time Zone: 0.00 (EUW)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-2.2	-0.7	-5.0	2.5	-0.7	-3.7	2.8	0.9	26.7	6.9	23.0	7.5	2.6	50	0.781

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	6.1	20.3	16.2	18.6	15.4	17.5	14.7	17.0	19.2	16.1	17.8	15.3	16.7	6.1	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.0	11.4	18.0	15.2	10.8	16.9	14.6	10.4	16.1	47.8	19.3	45.1	17.9	43.0	16.9	22.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 19.0	15.5	13.4	DB	-4.6	23.9	1.6	1.9	-5.8	25.3	-6.7	26.4	-7.6	27.4	-8.7	28.8	(4)
(5)			WB	-5.2	19.0	1.6	1.3	-6.3	19.9	-7.3	20.7	-8.1	21.4	-9.3	22.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	9.4	5.6	5.3	6.1	8.0	10.2	12.5	13.9	13.9	12.7	10.4	7.7	6.1		
	DBStd	3.92	2.52	2.46	2.37	2.36	2.53	1.99	1.96	1.67	2.01	2.42	2.62	2.82		
	HDD10.0	710	136	132	121	68	27	3	0	0	3	25	75	121		
	HDD18.3	3263	394	364	378	310	253	175	138	139	170	246	318	378		
	CDD10.0	489	0	1	1	8	32	78	122	120	83	37	6	2		
	CDD18.3	2	0	0	0	0	0	0	1	1	0	0	0	0		
	CDH23.3	6	0	0	0	0	1	1	3	1	0	0	0	0		
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0		
(14) Wind	WSAvg	6.5	7.9	7.1	7.0	6.6	6.3	5.3	4.9	5.3	6.0	6.8	7.2	7.3		
(15) Precipitation	PrecAvg	1314	148	92	114	77	59	71	88	103	129	152	145	143		
	PrecMax	1721	234	230	221	167	156	144	167	218	232	290	231	252		
	PrecMin	1035	25	2	27	8	11	31	30	24	26	43	50	57		
	PrecStd	141	54	47	43	39	32	27	33	48	47	54	45	50		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.6	11.5	13.4	17.3	21.8	21.5	23.4	21.2	18.7	16.2	14.0	12.6		
		MCWB	10.5	10.1	10.1	11.6	15.2	16.6	18.5	17.6	16.2	14.2	12.9	11.5		
	2%	DB	10.6	10.4	11.6	14.8	18.3	19.1	20.3	18.7	17.3	15.1	13.0	11.7		
		MCWB	9.9	9.4	9.5	10.6	13.4	15.2	16.9	15.9	15.2	13.5	11.9	10.8		
	5%	DB	10.0	9.7	10.5	12.9	16.0	17.4	18.5	17.5	16.4	14.4	12.2	11.0		
		MCWB	9.4	8.8	8.8	9.8	12.1	14.2	15.7	15.2	14.6	13.0	11.2	10.1		
	10%	DB	9.4	9.0	9.7	11.6	14.3	16.1	17.2	16.6	15.6	13.7	11.6	10.2		
		MCWB	8.7	8.0	8.2	9.0	11.1	13.3	14.7	14.5	14.1	12.4	10.5	9.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.7	10.6	11.2	12.7	15.7	17.2	18.7	18.0	16.9	14.8	13.2	11.7		
		MCDB	11.1	11.0	12.0	14.9	20.2	20.7	22.3	20.6	17.8	15.5	13.6	12.2		
	2%	WB	10.1	9.7	10.1	11.3	14.2	15.7	17.2	16.6	15.8	14.1	12.2	11.0		
		MCDB	10.4	10.1	10.9	13.8	17.6	18.2	19.7	18.3	16.7	14.7	12.7	11.6		
	5%	WB	9.6	9.0	9.2	10.3	12.6	14.6	16.2	15.7	15.1	13.4	11.5	10.2		
		MCDB	9.9	9.6	10.1	12.3	15.0	16.8	18.0	16.9	15.9	14.1	12.1	10.9		
	10%	WB	8.8	8.2	8.5	9.6	11.6	13.7	15.2	15.1	14.5	12.7	10.6	9.4		
		MCDB	9.3	8.8	9.5	11.2	13.5	15.6	16.7	16.0	15.3	13.6	11.4	10.1		
(35) Mean Daily Temperature Range	5% DB	MDBR	4.3	4.9	5.7	6.5	6.7	6.4	6.1	5.6	5.4	5.0	4.8	4.7		
		MCDBR	4.0	4.5	6.0	7.9	8.8	8.5	7.8	6.6	5.6	4.7	4.5	4.4		
	5% WB	MCWBR	4.0	4.1	4.7	5.0	5.4	5.4	5.1	4.5	4.3	4.0	4.5	4.5		
		MCWBR	3.8	4.2	5.1	6.7	7.6	7.5	6.9	5.7	5.1	4.3	4.5	4.4		
(40) Clear-Sky Solar Irradiance	taub	taub	0.345	0.353	0.372	0.384	0.388	0.380	0.382	0.369	0.366	0.350	0.335	0.332		
		taud	2.197	2.291	2.306	2.321	2.361	2.422	2.420	2.472	2.463	2.416	2.272	2.171		
	Ebn at Noon	Ebn at Noon	536	703	787	835	854	866	855	846	793	706	551	450		
		Edh at Noon	61	79	98	111	113	107	106	94	84	69	55	49		
(44) All-Sky Solar Radiation	RadAvg	RadAvg	0.47	1.16	2.28	3.81	4.93	4.98	4.66	3.76	2.54	1.38	0.61	0.34		
	RadStd	RadStd	0.05	0.11	0.17	0.26	0.42	0.49	0.35	0.29	0.16	0.14	0.06	0.04		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (41 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page